

Ammar Ahmed Al-Sayed Al-Nady

AI & Machine Learning Engineer | Data Scientist

Damietta, Egypt

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SUMMARY

AI & Machine Learning Engineer and Computer Science student with hands-on experience in Machine Learning, Data Science, Deep Learning, NLP, LLMs, and RAG-based applications. Experienced in developing ML models, performing data preprocessing and feature engineering, and building intelligent NLP systems using Transformers, Hugging Face, embeddings, semantic search, and query routing. Team Leader in the DEPI Customer Churn project and contributor to an NTI RAG-based research assistant, with a strong focus on practical AI development and continuous learning.

EDUCATION

Bachelor's degree in computers and Artificial Intelligence — Information Technology Department 2023 – 2027
Damietta University

WORK EXPERIENCE / INTERNSHIPS

Digital Egypt Pioneers Initiative (DEPI) Nov 2025 – Jul 2026

AI & Data Science — Microsoft Machine Learning Engineer Track

Training covered:

Python Programming & Data Science, Data Preprocessing & Visualization, Machine Learning, Deep Learning, Natural Language Processing with Attention Models, Computer Vision & Transfer Learning, Prompt Engineering, Statistics & calculus & Linear Algebra Fundamentals, Azure AI, MLOps & MLflow, Hugging Face, Capstone Project

Role: Team Leader — Customer Churn Prediction Project

National Telecommunication Institute (NTI) Aug 2026 – Sep 2026

Natural Language Processing (NLP) Track

Focused on:

Natural Language Processing, Transformers & Attention, LLMs, RAG, Prompt Engineering, Embeddings & Semantic Search, Query Classification & Routing, AI Agents, Web Search, NLP Project Development

Project: Paper Research Assistant

Technical Skills

Programming: Python, SQL, JavaScript

Data Science: NumPy, Pandas, Matplotlib, Seaborn, Exploratory Data Analysis (EDA), Data Cleaning & Preprocessing, Statistical Analysis, Hypothesis Testing, Feature Engineering, Feature Selection, Data Visualization

Machine Learning: Scikit-learn, XGBoost, Classification & Regression, Random Forest, SVM, Decision Trees, KNN, Naive Bayes, Clustering, Ensemble Learning, Cross-Validation, Hyperparameter Tuning, Model Evaluation, Imbalanced Classification

Deep Learning: TensorFlow, PyTorch, Keras, Neural Networks / ANN, CNN, RNN, LSTM, GRU, Transformers, Attention Mechanisms, Transfer Learning, Embeddings

NLP, LLMs & RAG: Natural Language Processing (NLP), Text Preprocessing, Text Classification, Sentiment Analysis, Named Entity Recognition (NER), Text Similarity, Transformers, Hugging Face, LLMs, Prompt Engineering, Retrieval-Augmented Generation (RAG), Document Chunking, Embeddings, Vector Search, Semantic Search, Query Classification, Query Routing, AI Agents, Web Search

AI / LLM Frameworks: LangChain (Basic), LangGraph (Basic), LlamaIndex (Basic), Hugging Face, Gemini API

Computer Vision: OpenCV, Image Preprocessing, Image Classification, CNNs, Object Detection, Face Recognition, Transfer Learning, YOLO

AI Engineering & MLOps: FastAPI (Basic), MLflow (Basic), Git, GitHub, Azure, Azure Machine Learning, Hugging Face Spaces

Databases: SQL, MySQL, SQL Server, Database Design, Data Aggregation, JOINS, Subqueries, CTEs

Additional Knowledge: Linux, CCNA 200-301, HTML, CSS

PROJECTS

Paper Research Assistant	NTI NLP Track Team Project	Sep 2026
An Intelligent RAG-Based Research Assistant with Topic Analysis and Web Search		
• Developed the Research Agent and query-routing components for an intelligent research assistant based on LLMs and RAG. • Implemented query classification and routing across RAG, Web Search, and Hybrid workflows. • Developed search query preparation logic to transform user requests into effective web-search queries. • Integrated DuckDuckGo for web search with OpenAlex as a fallback source. • Worked with Qwen/Qwen2.5-72B-Instruct and Hugging Face models within the project pipeline. • Worked with semantic retrieval using all-MiniLM-L6-v2 embeddings, chunking, top-k retrieval, and context construction. • Contributed to testing and validation of the project, with 90 tests passed and 1 skipped in the latest test run.		
Technologies: Python, NLP, LLMs, Qwen, Hugging Face, RAG, Embeddings, Semantic Search, Query Classification, Query Routing, DuckDuckGo, OpenAlex		

Customer Churn Prediction & Automated Retraining System	Team Leader	Jul 2026
DEPI — AI & Data Science / Microsoft Machine Learning Engineer		
• Led the team in developing a customer churn prediction system using a telecom dataset containing approximately 100K records. • Performed data preprocessing, exploratory analysis, feature engineering, model development, and evaluation. • Developed and trained an XGBoost classification model for customer churn prediction and handled class imbalance during training. • Achieved 0.9523 ROC-AUC and 0.84 recall for the churn class. • Implemented probability-based churn prediction and risk categorization into Low, Medium, and High Risk. • Worked on feature contribution analysis to help explain factors associated with individual churn predictions. • Worked with Azure Machine Learning as part of the project workflow. • Collaborated with the team on integrating the ML model into the final dashboard and automated retraining workflow.		
Technologies: Python, Pandas, NumPy, Scikit-learn, XGBoost, Machine Learning, Feature Engineering, Imbalanced Classification, Model Evaluation, Azure ML		

CERTIFICATIONS

AI & Data Science — Microsoft Machine Learning Engineer	Nov 2025 – Jul 2026
Digital Egypt Pioneers Initiative (DEPI), MCIT	
Natural Language Processing (NLP)	Aug 2026 – Sep 2026
National Telecommunication Institute (NTI) & ITIDA	
Introduction to Deep Learning	Mar 2026
Information Technology Institute (ITI) / Mahara-Tech	

LANGUAGES

Arabic (Native)
English (B1)